

Employee Management

Overview

An Employee Management application built with ASP.NET Core MVC, .NET Web API, and SQL Server. Includes an NUnit test project for integration and unit testing.

It allows users to perform CRUD operations on employee data through a web interface and API endpoints.

Project Structure

- ❖ EmployeeManagement -Solution
 - EmployeeManagement – **Presentation Layer**
 - ◆ Employee Controllers – Handle UI Requests
 - ◆ Views
 - Index – Employee List Page
 - CreateEmployee – Create /Edit (Employee creation and listing are handled asynchronously (no page reload))
 - ◆ Appsettings.json – Db Configuration for web
 - EmployeeManagementAPI - **API Layer**
 - Employee Controllers – API Endpoints
 - EmployeeData - **Data Access Layer**
 - EmployeeDbContext
 - ◆ EmployeeDbContext – Database Context
 - EmployeeModel – Table models
 - ResponseModel – Request response model
 - EmployeeServices - **Business Logic Layer**
 - Employee
 - ◆ EmployeeService – Employee CRUD operation service/Interface
 - ◆ HelperFunctions – Validation helper
 - EmployeeTest - **Testing Layer**
 - API -API integration Test Setup
 - Web - WEB integration Test Setup

Technologies Used

- ASP.NET Core MVC
- .NET Web API
- SQL Server
- Entity Framework Core
- AJAX
- Bootstrap
- NUnit (Testing)
- Moq (Mocking)
- ASP.NET Core MVC Testing

Run the Application Locally

Prerequisites:

- Visual Studio 2022/2026
- SQL Server

Startup Configuration

Set Multiple Startup Projects:

- EmployeeManagement
- EmployeeManagementAPI

Run the Application

- Start solution

URLs:

- Web Application (Frontend)
https://localhost:7211
 - Opens Employee CRUD UI
- API Base URL
https://localhost:7224
 - Use this in Postman to test APIs

API Testing (Postman)

A sample Postman collection is included in: /PostmanCollection

Sample API Request

URL: <https://localhost:7224/api/Employee/CreateEmployee>

Method: Post

Body(Json):

```
{  
  "employeeId": 0,  
  "employeeName": "finny johnn",  
  "designation": "Software Developer",  
  "emailAddress": "john@test.com",  
  "mobilePhone": "65545454",  
  "dateOfBirth": "1996-06-15",  
  "nationality": "indian",  
  "isActive": true  
}
```

Database Setup

- The project uses SQL Server as the database.
- Entity Framework Core is used for database configuration and data access.

Database Script

- The database creation and table scripts are included in the solution under:
/Db

Running Tests

Open Test Explorer in Visual Studio:

Test → Test Explorer

- Run all tests before starting the application
- The test project includes:
 - API integration tests
 - Web integration tests

Results will be displayed directly in Test Explorer

